

AgroScan

Dataset References

CSE400 viva notes — sources used to train and evaluate the vision models
and the supporting tabular / chat corpora

September 2026

How to use this list

If a panel member asks “which datasets did you use?”, start with the three vision trees, then name the papers.

AgroScan does **not** train one model on one dataset. Images from several public collections are merged, labels are rewritten to the **Crop___Disease** form, and then split into three folders:

- `Datasets/leaf_gate/` — leaf versus non-leaf (Level 1)
- `Datasets/leaf_type/` — 27 crop classes (Level 2)
- `Datasets/train|valid|test/` — about 123 disease classes (Level 3)

Disease training default is `combined` (`agrosan/config.py`). Loaders: `agrosan/data.py`. Label alignment: `agrosan/label_map.py`.

One sentence: “The CNNs are trained on PlantVillage, an augmented PlantVillage release, PlantDoc, and Bangladesh crop-leaf sets; ImageNet is only the transfer-learning start.”

Chapter 1

Vision datasets (cite these first)

Table 1.1: Datasets that feed the three AgroScan CNNs.

Dataset	Used in	What AgroScan takes
PlantVillage [1, 2]	L2 crop, L3 disease; L1 leaf positives	Lab-style healthy and diseased leaves; canonical <code>Crop___Disease</code> names
New Plant Diseases Dataset (Augmented) [3]	L3 train/valid (main tree)	Offline-augmented PlantVillage, ~38 classes, ~87k images
PlantDoc [4]	L3 (field-style)	2,598 internet/field images, 13 species, 27 classes
BD multi-crop disease [5]	L3 Bangladesh classes	123,588 images, 94 classes, 13 crops (gated Hugging Face set)
Bangladeshi leaf zip [6]	L3 extra local classes	Bean, Corn, Red-Amaranth from <code>Leaf Disease detection Dataset.zip</code>
Mango Dataset 2012 [7]	L2 crop only	Variety photos; not copied as extra disease classes
Non-leaf / invalid photos	L1 negatives	Reject selfies and random objects at the gate
ImageNet [8, 9]	all CNNs	Pretrained weights only (EfficientNet-B3, ResNet-50, DenseNet-121, MobileNetV3)

1.1 PlantVillage

Primary academic source for plant-leaf images. Hughes and Salathé released the repository [1]. Mohanty, Hughes and Salathé then trained CNNs on it and reported high lab accuracy [2]. AgroScan keeps their `Crop___Disease` naming so crop-masking at inference is well defined.

Say in viva: PlantVillage is mostly photographed in controlled light. That is why accuracy on the test split is high, and why we still use a leaf gate and a 90% crop-confirm step on phone photos.

Link: <https://arxiv.org/abs/1511.08060> <https://github.com/spMohanty/PlantVillage-Dataset>

1.2 New Plant Diseases Dataset (Augmented)

Kaggle redistribution by Samir Bhattarai [3]. It is PlantVillage with offline augmentation (~87k RGB images, 38 classes, 80/20 train/valid). In this repo the live disease splits follow that layout: `Datasets/train` and `Datasets/valid`. Config path: `plantvillage_augmented`

/ New Plant Diseases Dataset(Augmented).

Link: <https://www.kaggle.com/datasets/vipooooool/new-plant-diseases-dataset>

1.3 PlantDoc

Singh et al., ACM CoDS-COMAD 2020 [4]. Field-style and web images (2,598 images, 13 species, up to 17 diseases / 27 classes including healthy). Added so the disease models see cluttered backgrounds, not only lab leaves. Folder names are mapped to PlantVillage labels in `label_map.py` (PLANTDOC_TO_CANONICAL).

Link: <https://arxiv.org/abs/1911.10317> DOI: 10.1145/3371158.3371196

1.4 Bangladesh Multi-Crop Disease Classification Dataset

Hugging Face set `Saon110/bd-crop-vegetable-plant-disease-dataset` [5]. Gated, CC BY-NC-SA 4.0. Card reports 123,588 images, 94 classes, 13 crops (banana, cauliflower, corn, cotton, guava, jute, mango, papaya, potato, rice, sugarcane, tea, tomato, wheat). Downloaded by `agrosan/prepare_bd_dataset.py`, exported under `Datasets/vision/disease/bd_crop_disease/`.

Link: <https://huggingface.co/datasets/Saon110/bd-crop-vegetable-plant-disease-dataset>

1.5 Bangladeshi Leaf Disease Detection Dataset

Local zip used by `agrosan/bd_leaf_dataset.py`: `Leaf Disease detection Dataset.zip` under `Datasets/sources/bangladeshi_leaf_disease/` [6]. Classes in code: Bean, Corn, Red-Amaranth. Extracted once to `Datasets/vision/disease/bd_leaf_disease/`.

1.6 Mango variety set (2012)

Folder `mangifera2012/Mango_Dataset_2012` [7]. Labels become `Mango___Variety_....`. Used to help Level 2 recognise mango. They are **not** extra Level 3 disease classes (same rule as the P3 report).

1.7 What is not a vision training set

The advice pack under `data/agrosan/` is written cards (DAE/BARI/BRRI-style sources), not CNN labels. Chat RAG reads that pack. KrishokChat and crop CSVs below are optional text/tabular data, not leaf photos.

Chapter 2

Optional tabular and chat corpora

Table 2.1: Non-image datasets in the repo. They do not train the disease CNNs.

Dataset	Used in	Note
KrishokChat-145k [10]	optional LoRA / QA build	Bengali agri Q&A; <code>llm/download_datasets.py</code>
Crop recommendation CSV [11]	farm planner	<code>Datasets/tabular/crops/final_crops_data.csv</code>
Bangladesh crop 50k CSV [12]	farm planner / QA	<code>Bangladesh_Crop_Dataset-50k.csv</code>

Chapter 3

Merged training numbers (for the same question)

After merge and cleaning, the operational disease split used in the report is on the order of 241,000 / 30,000 / 30,000 images (train / validation / test), about 123 classes, 27 crops. Exact class lists live in the checkpoints and in `web/offline-pack.json`.

Chapter 4

References

Bibliography

- [1] D. P. Hughes and M. Salathé, “An open access repository of images on plant health to enable the development of mobile disease diagnostics,” arXiv:1511.08060, 2015. <https://arxiv.org/abs/1511.08060>
- [2] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016, doi: 10.3389/fpls.2016.01419. <https://doi.org/10.3389/fpls.2016.01419>
- [3] S. Bhattarai, “New Plant Diseases Dataset (Augmented),” Kaggle, 2018. <https://www.kaggle.com/datasets/vipoooool/new-plant-diseases-dataset>
- [4] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, “PlantDoc: A dataset for visual plant disease detection,” in *Proc. 7th ACM IKDD CoDS and 25th COMAD*, Hyderabad, India, 2020, pp. 249–253, doi: 10.1145/3371158.3371196. Also arXiv:1911.10317. <https://arxiv.org/abs/1911.10317>
- [5] S. C. Saon, “Bangladesh Multi-Crop Disease Classification Dataset,” Hugging Face, 2025. <https://huggingface.co/datasets/Saon110/bd-crop-vegetable-plant-disease-dataset>
- [6] “Bangladeshi Leaf Disease Detection Dataset,” local archive `Leaf Disease detection Dataset.zip` (Bean, Corn, Red-Amaranth), used via `agrosan/bd_leaf_dataset.py`.
- [7] “Mango Dataset 2012” (`mangifera2012`), variety photographs used only in AgroScan Level 2 crop identification.
- [8] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” in *Proc. IEEE CVPR*, 2009, pp. 248–255.
- [9] O. Russakovsky *et al.*, “ImageNet large scale visual recognition challenge,” *International Journal of Computer Vision*, vol. 115, pp. 211–252, 2015.
- [10] RaiyanKhaan, “KrishokChat-145k,” Hugging Face dataset. <https://huggingface.co/datasets/RaiyanKhaan/KrishokChat-145k>
- [11] “Appropriate Crop Recommendation Dataset for Cultivators,” local CSV `final_crops_data.csv` under `Datasets/tabular/crops/`.
- [12] “Bangladesh Crop Dataset-50k,” local CSV `Bangladesh_Crop_Dataset-50k.csv` under `Datasets/tabular/crops/`.

Viva cheat sheet

1. **Main vision papers:** Hughes and Salathé (PlantVillage, 2015); Mohanty et al. (2016); Singh et al. (PlantDoc, 2020).
2. **Bangladesh images:** Saon Hugging Face set (2025) plus the local Bean/Corn/Red-Amaranth zip.
3. **Augmented Kaggle copy:** Bhattarai, New Plant Diseases Dataset.
4. **Weights start from:** ImageNet.
5. **Advice is not a dataset:** pack JSON, not CNN labels.